

Mutual Fund Analytics Capstone Project

Final Report

Project Overview: End-to-end mutual fund analytics solution covering ETL, SQLite database, EDA, performance analytics, Power BI dashboard and advanced analytics.

Objectives

- ETL pipeline
- SQLite integration
- Exploratory Data Analysis
- Performance metrics (Sharpe, Beta, Alpha, Sortino, VaR, CVaR)
- Interactive Power BI Dashboard
- Advanced Analytics and Fund Recommendation

Datasets

NAV History, Scheme Performance, Investor Transactions and Portfolio Holdings datasets were cleaned, transformed and analyzed.

Methodology

Raw CSV files were ingested using Python, cleaned with Pandas, stored in SQLite, analyzed through Jupyter notebooks and visualized using Power BI.

Major Analyses

EDA, CAGR, Sharpe Ratio, Alpha, Beta, Sortino Ratio, Maximum Drawdown, Historical VaR, CVaR, Rolling 90-Day Sharpe Ratio, Investor Cohort Analysis, SIP Continuity Analysis, Sector HHI Concentration and Risk-based Recommendation.

Dashboard

Interactive dashboard includes fund performance, investor analytics, risk metrics and category comparisons with slicers and KPIs.

Key Business Insights

1. Risk-adjusted returns differ significantly across schemes.
2. Rolling Sharpe highlights changing market conditions.
3. Investor cohorts reveal long-term investment behavior.
4. SIP continuity identifies at-risk investors.
5. HHI measures portfolio concentration.

Technology Stack

Python, Pandas, NumPy, Matplotlib, Seaborn, SQLite, SQL, Power BI, GitHub.

Conclusion

The project successfully demonstrates an end-to-end mutual fund analytics platform capable of supporting data-driven investment decisions.